

RGB-T Object Detection With Failure Scenarios

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Abstract—Currently, RGB-thermal (RGB-T) object detection algorithms have demonstrated excellent performance, but issues such as modality failure caused by fog, strong light, sensor damage, and other conditions can significantly impact the detector's performance. This article proposes a multimodal object detection method named diffusion enhanced object detection network (DENet), aiming to address modality failure problems caused by nonroutine environments, sensor anomalies, and other factors, while suppressing redundant information in multimodal data to improve model accuracy. First, we design a multidimensional incremental information generation module based on a diffusion model, which reconstructs the unstable information of RGB-T images through the reverse diffusion process using the original fusion feature map. To further address the issue of redundant information in existing RGB-T object detection models, a redundant information suppression module is introduced, minimizing cross-modal redundant information based on mutual information and contrastive loss. Finally, a kernel similarity-aware illumination module (KSIM) is introduced to dynamically adjust the weighting of RGB and thermal features by incorporating both illumination intensity and the similarity between modalities. KSIM can fine-tune the contribution of each modality during fusion, ensuring a more precise balance that improves recognition performance across diverse conditions. Experimental results on the DroneVehicle and VEDAI datasets show that DENet performs outstandingly in multimodal object detection tasks, effectively improving detection accuracy and reducing the impact of modality failure on performance.

Index Terms—Diffusion model, kernel method, multimodal remote sensing, object detection, RGB-thermal (RGB-T) images.

I. INTRODUCTION

FROM early traditional techniques to today's deep learning-based approaches, the development of object detection algorithms has spanned over twenty years. Currently, single-modal object detection algorithms based on RGB images have made significant progress, with detection accuracy continuously

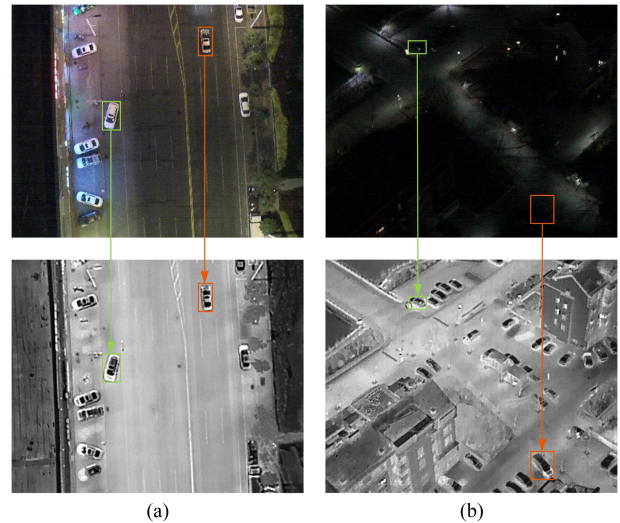


Fig. 1. First row is RGB images, and the second row is the corresponding infrared image. (a) Displays a scene with good lighting conditions, where the object is clear and visible, and training with only RGB images can achieve good results. (b) Scene with poor lighting conditions, in which the object in the RGB image is no longer visible and the model needs to be trained in conjunction with infrared images.

improving. These algorithms have transitioned from theoretical research to practical applications in various real-life scenarios, such as remote sensing [1], [2], autonomous driving [3], and urban security [4], achieving substantial success. However, when faced with challenges such as abnormal lighting and extreme environments, these RGB image-based object detection algorithms often fail to perform effectively. Consequently, researchers have explored training object detection models using infrared images to maintain robust detection performance under extreme conditions [5], [6], as shown in Fig. 1. Considering the complementary features of texture and lighting between RGB images and infrared images, researchers have increasingly focused on fusing RGB and infrared images for object detection to further enhance accuracy in extreme environments.

Despite numerous technical solutions proposed by researchers for fusing RGB and infrared images for object detection, RGB-T object detection models still face two significant issues.

- 1) **Modality Failure:** In certain situations, visible light or infrared sensors may fail due to environmental factors such as strong light, smoke, or darkness, leading to data loss or quality degradation, as shown in Fig. 2, which adversely affects the performance of the fusion model. If the model overly relies on information from one modality, the overall

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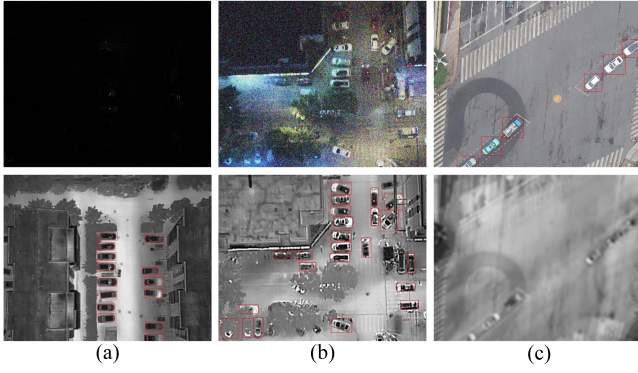


Fig. 2. First row shows RGB images, and the second row shows the corresponding infrared images. (a) Illustrates a scenario of RGB modality failure due to extreme lighting conditions. (b) Displays a RGB modality failure scenario caused by noise. (c) Shows a infrared modality failure scenario where the image is blurred due to lens defocus.

detection performance will significantly decrease when that modality fails;

- 2) **Information Redundancy:** During the fusion of RGB and infrared images, the similar feature information contained in both modalities can lead to redundancy if not properly handled. This redundancy increases computational burden without significantly improving detection performance. In addition, noise and errors from different sensors can be amplified during the fusion process, further impacting the accuracy of detection results.

Recently, some researchers have proposed methods to address the issue of multimodal image information redundancy in RGB-T object detection models. Wang et al. [6] introduced a redundant information suppression network, which reduces redundancy between RGB and infrared images by designing a mutual information minimization module, thereby enhancing the fusion of complementary multimodal information. Wang et al. [7] proposed an uncertainty-aware module, combined with a two-stream backbone network and an attention mechanism, to design a cross-modality feature fusion module that fully extracts shared and complementary information while suppressing redundant information, achieving comprehensive extraction and fusion of different modality features. Hu et al. [8] designed a patchwise cross-attention mechanism, utilizing multiscale features from different modalities for spatial dimension fusion, reducing redundant information and enhancing the extraction and fusion of effective information.

Regarding the issue of modality failure, Mees et al. [9] proposed an adaptive fusion method that addresses modality failure by online learning of the weights of different sensor modalities' predictions. This method is based on a convolutional neural network expert mixture model, combining various features such as appearance and motion, enabling the model to adaptively adjust and improve object detection performance even under extreme lighting changes and severe camera motion blur. Bijelic et al. [10] introduced a deep fusion network that solves the sensor modality failure problem through adaptive feature fusion.

Currently, effective solutions to the modality failure problem in the field of RGB-T object detection are still relatively scarce. Most existing solutions primarily address scenarios involving

abnormal lighting and motion blur. However, they tend to fail when faced with issues such as blur and distortion causing by sensor aging and malfunctions. In this article, we innovatively propose a diffusion model-based approach to address the modality failure problem. Simultaneously, we introduce and improve the mutual information mechanism based on information theory to eliminate redundant information between modalities and enhance feature quality. The main contributions of our work can be summarized as follows.

- 1) We designed a multidimensional incremental information generation module based on a diffusion model to address the issue of modality collapse caused by the instability of RGB-T image information quality. This module uses the diffusion model to reconstruct unstable information in RGB-T images, fully exploring the multidimensional incremental information of RGB-T images;
- 2) To address the issue of redundant information in RGB-T images weakening the model's ability to fuse multidimensional complementary information, this article proposes a redundant information suppression module (RISM) optimized through a combination of mutual information and contrastive learning. The network optimizes the parameters of the feature extraction network by incorporating mutual information of feature distributions and contrastive loss during training, effectively reducing redundant information in RGB-T image features and enhancing the model's ability to fuse multidimensional complementary information;
- 3) This article designs a kernel similarity-aware illumination module (KSIM) which introduces a dynamic mechanism that adjusts the contribution of RGB and thermal features based on both illumination and modality similarity. This innovation ensures more precise and adaptive feature fusion, enhancing robustness across varying conditions, and overcoming the limitations of fixed illumination-based strategies.

The rest of this article is organized as follows. In Section II, we review the technologies related to our proposed method. In Section III, we describe our proposed RGB-T object detection algorithm in detail. In Section IV, we present and analyze the experimental results of the algorithm on two datasets. In Section V, we discuss the model's performance in scenarios of modality failure and conduct further comparative experiments based on the datasets processed by image degradation. Finally, Section VI concludes this article.

II. RELATED WORK

A. Multimodal Object Detection

RGB images contain rich texture and color information, while infrared images are valued for their ability to clearly present target contours. Each of these image types has its own unique informational advantages, and their combined use in the field of object detection has become a research focus in recent years [11], [12]. Many researchers have conducted in-depth studies from different research perspectives on how to effectively integrate the complementary information of RGB and infrared images.

RGB-T object detection models can be categorized into three major types based on the fusion method of RGB and infrared images: pixel-level fusion models, feature-level fusion models, and decision-level fusion models.

In pixel-level fusion, RGB images and infrared images are first fused at the image level, and the fused image is then used as input for object detection. Common fusion methods include weighted averaging, channel concatenation, and channel replacement [13], [14]. Wagner et al. [15] combined the three channels of RGB images with the infrared image channel to form a four-channel image, which was then input into the object detection network to achieve multimodal object detection. However, experimental results showed that simply merging the channels without preprocessing did not produce good results. In 2023, Zhang et al. [16] proposed an attention-guided pixel-level fusion strategy. The authors extracted attention vectors from RGB images and infrared images, applied them to the original images, and merged the enhanced images with the original images to obtain the fused image, which was then used for subsequent feature extraction operations. Pixel-level fusion strategies directly utilize the raw information from both modalities, allowing for the preservation of more detailed information in the early stages. However, since pixel-level fusion often overlooks the feature differences between the two modalities in different scenarios, it can easily introduce redundant information or noise, making it difficult to handle multimodal feature alignment in complex scenes.

Currently, most research on RGB-T object detection focuses on feature-level fusion strategies. Feature-level fusion methods involve sending RGB and infrared images into separate feature extraction networks and then fusing the features at intermediate or deep layers. Benefiting from the development of attention mechanisms, most researchers now adopt attention-based adaptive fusion of different modality features. By integrating various attention mechanisms into the network, these methods extract and fuse the key features from different modalities, aiming to achieve the fusion of multidimensional complementary information from RGB-T images. For example, Yang et al. [17] employed attention mechanisms for cross-modal fusion. They used a bidirectional multistage attention strategy to progressively extract and enhance RGB and infrared features while preserving modality-specific features. Zhang et al. [18] designed a cross-modal attention module to enhance and complement the differential features between modalities. At the same time, model used an intramodality attention module to enhance the foreground features of each modality and fused both cross-modal and intramodality features for object detection. Wu et al. [19] designed a multimodal adaptive feature fusion module based on attention weights. This module enhances high-weight features while compressing low-weight redundant features during fusion. The weights are computed through a soft pooling channel attention mechanism calculating from RGB and infrared modalities. Wang et al. [7] designed a dual-stream backbone network based on attention mechanisms, incorporating a cross-modal attention feature fusion module to fully extract shared and complementary features. They also employed a three-branch feature enhancement network to separately learn features from three different

stages of network propagation. Xu et al. [20] used both channel attention mechanisms and spatial attention mechanisms to process RGB and infrared features, calculating weights across different dimensions to perform weighted operations on the RGB and infrared features. They also employed self-attention mechanisms to further enhance feature representations and complete the cross-modal feature fusion. Feature-level fusion can extract more distinctive deep features from different modalities and preserve the advantageous features of each modality during fusion. However, selecting the optimal fusion strategy poses certain challenges, as incorrect fusion strategies can lead to information loss or insufficient feature representation.

Decision-level fusion models perform independent object detection on RGB images and infrared images to obtain separate detection results. These results are then fused at the decision level to produce the final object detection results. For example, Li et al. [21] used two independent networks to predict confidence scores and bounding box parameters from RGB images and infrared images, respectively. The final results were then obtained by fusing the confidence scores and bounding box parameters of different modalities based on weights generated by an illumination module. Chen et al. [22] proposed a probabilistic ensemble strategy that uses Bayesian rules to probabilistically fuse detection results from RGB and thermal images. Even when the condition of independence does not hold, this method effectively integrates outputs from multiple detectors and improves multimodal object detection performance. Li et al. [23] employed a dense fusion strategy to extract multilevel multispectral feature representations at the feature level. In addition, they used an extra confidence subnet to dynamically estimate the detection confidence for each modality. The final results were fused using Dempster's combination rule based on the adjusted confidence. However, this study did not address the issue of how modality failure and multidimensional redundant information in RGB-T images might affect the model's detection performance.

B. Modality Failure in Image Processing

In the field of multimodal object detection, the diversity of data collection environments and the differences in sensor characteristics lead to instability in modality information and the potential bias of the model toward a particular modality during optimization. This phenomenon not only impacts the model's generalization ability but also limits the effectiveness of multimodal fusion strategies. Therefore, exploring how to fully extract the incremental information from multimodal data and ensuring that the model balances the contribution of each modality during the optimization process has become a critical issue that needs to be addressed in this field.

Currently, the mainstream methods for addressing modality failure primarily fall into two categories. The first category is generation-based methods, which aim to directly predict and reconstruct missing modality information based on the available input modality data. For example, Tran et al. [24] proposed the cascaded residual autoencoder (CRA), which leverages residual connections within an autoencoder structure to recover corrupted data through an estimation function. Zhao

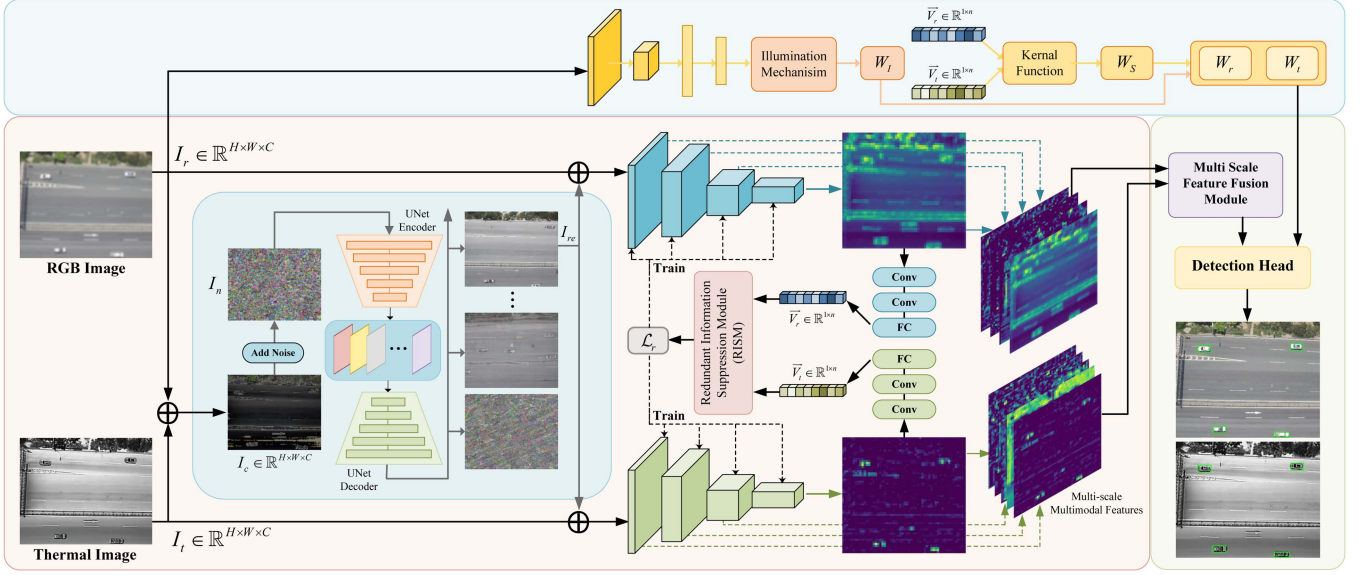


Fig. 3. Network architecture of DENet: (a) a diffusion enhancement module based on early-fused images to restore and enhance image features (the blue region following the input); (b) RISM based on dual-modality feature vectors to suppress information redundancy caused by diffusion enhancement and original features (the red block below the backbone); and (c) KSIM based on the illumination-aware module to balance the contributions of different modalities to the final detection result (the blue region at the top). The multiscale feature fusion module and detection head are adopted from the baseline model. $\oplus$ represents pixelwise weighted averaging.

et al. [25] developed the missing modality imagination network based on CRA and a cyclic consistency learning strategy. This network can reconstruct missing modality features given the representation of available modality information. In 2021, Ma et al. [26] designed a feature reconstruction network based on Bayesian metalearning, which can reconstruct missing modality features using the available modality information. Other similar methods include the conditional variational autoencoder [27], which learns structured output representations using deep conditional generative models, the conditional multimodal autoencoder [28], which applies variational methods for conditional multimodal deep learning, and the multimodal variational autoencoder [29].

The second category of methods focuses on multimodal joint representation learning, which seeks to represent multimodal data by exploring and integrating complementary information across different modalities. Han et al. [30] proposed a joint training model composed of two modality-specific encoders and a shared classifier, which combines audio and visual information into a joint representation to enhance the performance of emotion recognition in a single modality. In the same year, Pham et al. [31] introduced a sequential translation-based model that learns joint representations between a source modality and multiple target modalities by using the hidden vectors from the source modality's encoder. This approach improved emotion recognition performance for the source modality. Regarding the issue of modal failure, Javaloy et al. [32] in 2022 analyzed its causes, identifying gradient conflicts in certain parameters during model optimization as a key factor. To mitigate this problem, they proposed a training strategy for multimodal variational autoencoders, which employs existing multitask learning solutions to ensure an impartial optimization process that does

not favor any particular modality. Wang et al. [33] observed that the information quality from sensors can fluctuate significantly across samples. To address this, they proposed a method using uncertainty estimation and evidential fusion modules to handle information uncertainty, thereby enhancing the robustness and reliability of RGB-T semantic segmentation.

III. METHODOLOGY

A. Architecture

The overall structure of the diffusion enhanced object detection network (DENet) is shown in Fig. 3. DENet mainly consists of a diffusion enhancement module (DEM), RISM, KSIM, and a feature fusion module.

DENet takes the RGB-T image pair I_t and I_r as input data and first merges the images I_t and I_r to obtain the fused image I_c . The fused image I_c is then input into the DEM to reconstruct the information lost due to modality failure, while also extracting the incremental information from the RGB-T images, resulting in the incremental information image I_{re} . Subsequently, the original images and the incremental information image are used to obtain I'_t and I'_r , which are fed into a dual-stream backbone feature extraction network (ResNet 50) to extract the image pair features, resulting in feature maps F_t and F_r . The feature maps of the two modalities are then fed into the RISM to measure the similarity of the two modality features, further obtaining the redundancy information loss, aiming to enable the backbone to extract more unique features for each modality. The feature vectors extracted from the feature maps are also fed into the KSIM to obtain the modality information balance optimization weight W_s . This weight is used to adjust the illumination weight W_t generated by the illumination-aware submodule, balancing

the contributions of different modalities to the final detection result. Finally, the multiscale stable RGB-T features are input into the fusion and detection module to predict the location and category of the objects.

B. Diffusion Enhancement Module

Reconstructing the uncertainty information in RGB-T images and extracting the incremental information from RGB-T images are crucial for enhancing the model's ability to utilize the multidimensional information of RGB-T images. In this chapter, based on RGB-T images with unstable information quality, a diffusion model is used to reconstruct the damaged information in RGB-T images and extract the incremental information from them.

The RGB image I_r and the thermal infrared image I_t are merged to obtain the image I_c . The diffusion model, guided by the image I_c , reconstructs the information lost due to modality failure, resulting in the incremental information image I_{re} . Specifically, the image I_c undergoes diffusion processing, where different levels of noise are added to the image I_c , generating the noisy image I_n as follows:

$$I_n, \mathcal{N} = f\left(\frac{I_r + I_t}{2}\right) \quad (1)$$

where $f(\cdot)$ represents the diffusion processing operation. The diffusion model is then trained using the noisy image I_n and noise distribution N to fit the noise distribution N

$$\tilde{\mathcal{N}} = f_{dm}(I_n) \quad (2)$$

where $f_{dm}(\cdot)$ represents the diffusion model, and $\tilde{\mathcal{N}}$ represents the noise distribution predicted by the diffusion model. During the reverse diffusion process of the diffusion model, which reconstructs the unstable information of the RGB-T image through sampling operations, the images I_t and I_r are used for sampling to obtain the image I_{re} as follows:

$$I_{re} = \tilde{f}\left(\Phi\left(\frac{I_r + I_t}{2}\right)\right) \quad (3)$$

where Φ represents converting data into a standard normal distribution, and $\tilde{f}(\cdot)$ represents the inverse diffusion sampling operation.

C. Redundant Information Suppression Module

DEM aims to reconstruct modal-failure image information through diffusion processes and generate incremental information to compensate for modality failures. However, it may generate incorrect or unrelated information to the original modality under extreme circumstances. If not controlled, it could introduce interference or redundant features, thereby affecting the detection performance of the model. In addition, the nonlinear complexity of the feature information extracted from the original RGB-T images makes it challenging for the model to reveal the relationships between multidimensional information. Ideally, the model should fuse the complementary multidimensional information from RGB-T images. However, in practice, the

model unavoidably fuses redundant multidimensional information from RGB-T images during the undifferentiated fusion process. Therefore, this module aims to filter out the redundant multidimensional information of RGB-T images during the feature extraction stage to address the issue of incorporating redundant multidimensional information during the feature fusion stage. This module is an improved version based on the mutual information minimization module [6] we proposed in a previous work, and its structure is shown in Fig. 4.

In the original module, for the dual-modality features F_r and F_t extracted by the backbone network, two convolutional layers and fully connected layers are first used to map the feature maps F_r and F_t to two low-dimensional feature vectors $\vec{V}_r$ and $\vec{V}_t$. Further, based on these feature vectors, two information distributions $\mathcal{N}_r$ and $\mathcal{N}_t$ are created, and finally, the mutual information between the two information distributions is calculated. The calculation method is as follows:

$$M(\mathcal{N}_r, \mathcal{N}_t) = H(\mathcal{N}_r) + H(\mathcal{N}_t) - H(\mathcal{N}_r, \mathcal{N}_t) \quad (4)$$

where $M(\cdot)$ represents the mutual information between two vectors, $H(\cdot)$ represents the information entropy, and $H(\cdot, \cdot)$ represents the joint entropy of the two feature vectors.

At the network level, the goal is to minimize the mutual information, and considering the nonnegativity of joint entropy, when optimizing $H(\mathcal{N}_r)$ and $H(\mathcal{N}_t)$ to be minimal, it can be considered that the mutual information is minimized. Therefore, the mutual information optimization function is expressed as follows:

$$\mathcal{L}_m = H(\mathcal{N}_r) + H(\mathcal{N}_t). \quad (5)$$

In addition, the relationships between information entropy, cross entropy, and relative entropy can be represented as follows:

$$H(\mathcal{N}_r) = C(\mathcal{N}_t, \mathcal{N}_r) - D(\mathcal{N}_t || \mathcal{N}_r) \quad (6)$$

$$H(\mathcal{N}_t) = C(\mathcal{N}_r, \mathcal{N}_t) - D(\mathcal{N}_r || \mathcal{N}_t) \quad (7)$$

where $C(\cdot, \cdot)$ represents cross entropy and $D(\cdot, \cdot)$ represents relative entropy. By combining the above formulas, we can obtain the following function:

$$\mathcal{L}_m = C(\mathcal{N}_r, \mathcal{N}_t) + C(\mathcal{N}_t, \mathcal{N}_r) - D(\mathcal{N}_t || \mathcal{N}_r) - D(\mathcal{N}_r || \mathcal{N}_t). \quad (8)$$

To enhance the module's ability to measure the redundancy of multimodal image information and address the problem of potential interference and redundant features introduced by DEM under extreme conditions, we introduce contrastive loss in the mutual information minimization module. However, traditional contrastive learning typically relies on multiple positive and negative sample pairs for training. In the scenario of our proposed model, only two feature vectors, RGB and thermal features, are available. To address this issue of sample scarcity, we introduce virtual negative samples by adding random perturbations to one of the feature vectors, generating virtual negative samples with slight differences from the original features. The random perturbations were generated by Gaussian distribution. The adjusted

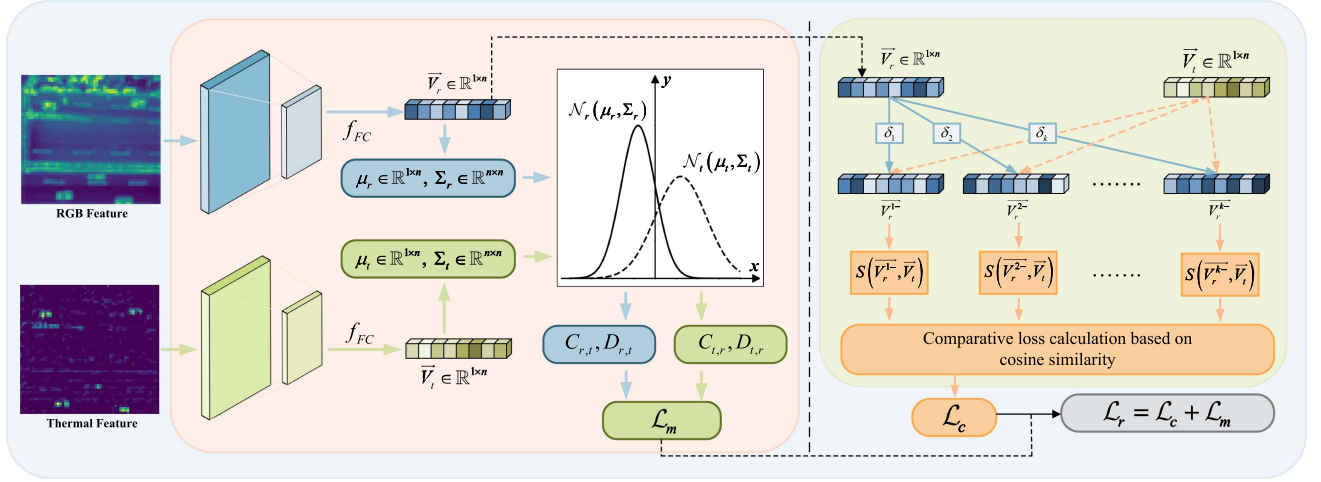


Fig. 4. Structure of the RISM. The feature maps of the two modalities are converted into vectors through convolutional and fully connected layers. The red area represents the loss generated by mutual information, the module generates a Gaussian distribution from the two vectors to calculate mutual information and further obtain the loss. The green area represents the loss generated by contrastive learning; the module adds perturbations to the vectors to generate several pseudo samples, thereby calculating the contrastive loss. Symbol $C_{r,t}$ in red area represents $C(\mathcal{N}_r, \mathcal{N}_t)$, $D_{r,t}$ represents $D(\mathcal{N}_r | \mathcal{N}_t)$, other similar symbols follow the same pattern.

contrastive loss is represented as follows:

$$\mathcal{L}_c = -\log \left(\frac{\exp(S(\vec{V}_r, \vec{V}_t))}{(\exp(S(\vec{V}_r, \vec{V}_t)) + \sum_k (\exp(S(\vec{V}_r^{k-}, \vec{V}_t)))} \right) \quad (9)$$

where $\vec{V}_r^{k-}$ is the k th pseudo sample generated by random perturbation and $S(\cdot, \cdot)$ represents cosine similarity. The reason for generating virtual samples by adding random perturbations to real feature vectors is that in the contrastive learning process of multimodal features, the quality of positive and negative samples directly affects the model's ability to measure feature similarity and differences. Since the virtual negative samples are generated from real feature vectors, the perturbed negative samples can effectively simulate the subtle differences that may exist in multimodal features. Compared to using completely random or unrelated negative samples, virtual negative samples can more finely approximate the variations in real-world multimodal feature environments, enabling the model to more accurately distinguish between similar and different features during the feature learning process.

The final mutual information optimization function can be expressed as follows:

$$\mathcal{L}_r = \mathcal{L}_m + \mathcal{L}_c \quad (10)$$

where $\mathcal{L}_r$ represents the amount of redundant information between RGB and infrared image features. By optimizing $\mathcal{L}$, the parameters of the feature extraction network are optimized, which in turn filters out the redundant information in the RGB-T image features. At the same time, it obtains RGB-T image features rich in complementary information. This approach alleviates the problem of being unable to distinguish between complementary and redundant information in the subsequent feature fusion stage.

D. Kernel Similarity-Aware Illumination Module

Although the RISM effectively reduces redundant information between RGB and thermal features, ensuring that each modality provides unique features, directly fusing results from two modalities during the fusion stage is unreasonable. In real-world scenarios, the similarity between features from different modalities can vary significantly under different conditions. Simple redundancy suppression cannot accurately control the contribution of each modality based on the actual feature characteristics. The model requires a mechanism that can dynamically adjust the contribution of different modality, ensuring that the fusion is more precise and reasonable. Most RGB-T object detection models control the weights of different modality during fusion through an illumination-aware module. However, in the proposed model, the illumination-aware module adjusts the weights of RGB and thermal results based on illumination intensity solely, which cannot flexibly address the differences between RGB and thermal modalities. When the RGB image has good illumination conditions, the model might incorrectly rely on illumination intensity to attach more importance to RGB results, neglecting the effectiveness of thermal results.

To address this issue, we introduce the KSIM, which dynamically adjusts the illumination weight distribution by measuring the similarity between RGB and thermal features. First, we can get two feature vectors $\vec{V}_r$ and $\vec{V}_t$ extracted from RISM through convolutional and fully connected layers. Next, a multikernel function $k(\vec{V}_r, \vec{V}_t)$ is used to measure the similarity between the low-dimensional feature vectors $\vec{V}_r$ and $\vec{V}_t$, resulting in a similarity weight W_S as follows:

$$W_S = k(\vec{V}_r, \vec{V}_t) = \exp \left(\frac{-\|\vec{V}_r - \vec{V}_t\|^2}{2\sigma^2} \right). \quad (11)$$

Then, the illumination-aware weight W_I generated by the illumination-aware module is adjusted through W_S to obtain the final multimodal features fusion weight. The calculation process is as follows:

$$W_r = \frac{W_I (1 + \beta(1 - W_S))}{1 + \beta} \quad (12)$$

$$W_t = 1 - W_r \quad (13)$$

where β represents the correction weight.

In scenarios where the modalities differ significantly, KSIM automatically reduces the reliance on RGB results and increases the contribution of thermal results. Conversely, when RGB and thermal feature similarity is high, the model can still rely on RGB results. This allows for a more precise response to cross-modal differences. Compared to the fixed strategy of a simple illumination-aware module, we introduce a more flexible dynamic adjustment mechanism by incorporating modality similarity measurements. This fine-tuned adjustment approach is not only suitable for various illumination conditions but also dynamically adapts to the changes in modality features, enabling the model to achieve optimal performance across a wider range of application scenarios.

IV. EXPERIMENT

In this section, we first introduce the DroneVehicle and VEDAI datasets used in the experiments. and then present the experimental results of our proposed algorithm alongside several comparison algorithms on these datasets. In addition, ablation experiments are conducted to demonstrate the effectiveness of the modules we proposed. Finally, we analyze the experimental results and draw conclusions.

A. Datasets

Currently, there are multiple datasets available for training RGB-T object detection models, such as VEDAI [34], KAIST [35], CVC-14 [36], DroneVehicle [37], and ODinMJ [38], etc. We conducted experiments using VEDAI and DroneVehicle datasets in remote sensing scenarios

DroneVehicle Remote Sensing Dataset: The DroneVehicle dataset [37] consists of 56 878 images collected by drones, with half being RGB images and the other half being the corresponding infrared images. The images cover various scenes such as city roads and highways, containing five types of objects: cars, trucks, buses, vans, and lorries. The dataset also includes scenarios with modality failures such as darkness and blurring. The original image resolution is 640×512 , and the authors added a 100-pixel blank area around the images, resulting in a final resolution of 840×712 . In our experiments, we processed the images by cropping them back to a size of 640×512 .

VEDAI Remote Sensing Dataset: The VEDAI Dataset [34] contains 1210 pairs of RGB-T images collected by drones, covering various types of vehicles such as cars, trucks, and tractors. The image resolution in this dataset is 512×512 , and the dataset includes several modality failure scenarios such as

TABLE I
EXPERIMENTAL RESULTS ON DRONEVEHICLE REMOTE SENSING DATASET

Methods	Modality			mAP(%)
	Visible	Thermal	Visible+Thermal	
YOLO v3 [39]	✓	×	×	56.40
Faster R-CNN [40]	✓	×	×	44.59
RetinaNet [41]	✓	×	×	57.37
DETR [42]	✓	×	×	52.67
YOLO v3 [39]	×	✓	×	59.32
Faster R-CNN [40]	×	✓	×	54.55
RetinaNet [41]	×	✓	×	60.93
DETR [42]	×	✓	×	56.15
UA-CMDet [37]	×	×	✓	74.07
MBNet [43]	×	×	✓	76.19
C ² Former [44]	×	×	✓	74.20
AFFCM [19]	×	×	✓	76.64
DENet (Ours)	×	×	✓	78.42

shadows and occlusions. Therefore, we will also use the VEDAI dataset for experiments.

B. Implement Details

All experiments based on our proposed model in this article were implemented using Tensorflow on NVIDIA 4090 GPU. The operating system is Ubuntu 20.04, and the CUDA version is 12.1. During the dataset preprocessing stage, we applied random color distortion and flipping to the images to increase data diversity. For the training parameters, the entire network was trained using the Adam optimizer for 25 epochs with an initial learning rate of 0.0001, and the learning rate was set to decrease by 10 times every 5 epochs. The batch size was set to 16. The correction weight β set to 0.4. For the perturbation vectors, we conducted experiments using different numbers of perturbation vectors. The experimental results showed that selecting six perturbation vectors for the experiment can achieve the best comprehensive performance of the model.

C. Experimental Results

In the DroneVehicle dataset, we compared our proposed DENet method with several single-modality object detection methods and advanced multimodal object detection methods. YOLO, faster R-CNN, DETR, and other single-modality object detection methods were trained and tested on RGB and thermal images separately. In addition, we selected five RGB-T object detection methods for comparison with our DENet method. The experimental results are shown in Table I. As indicated in the table, our proposed DENet achieved a mAP(%) score of 78.42%, which is 2.23% higher than the baseline model MBNet and outperforms the other RGB-T object detection models. Among the compared methods, the AFFCM method, proposed in 2023, achieved the best performance with a mAP(%) of 76.64%, while our method exceeded it by 1.78%. This is because, in the DroneVehicle dataset, some images contain scenes with haze, extreme lighting, and blurriness, which fall under the scope of modality failure. The DEM generates reliable enhanced images from the limited information in modality-failed images by fusing RGB

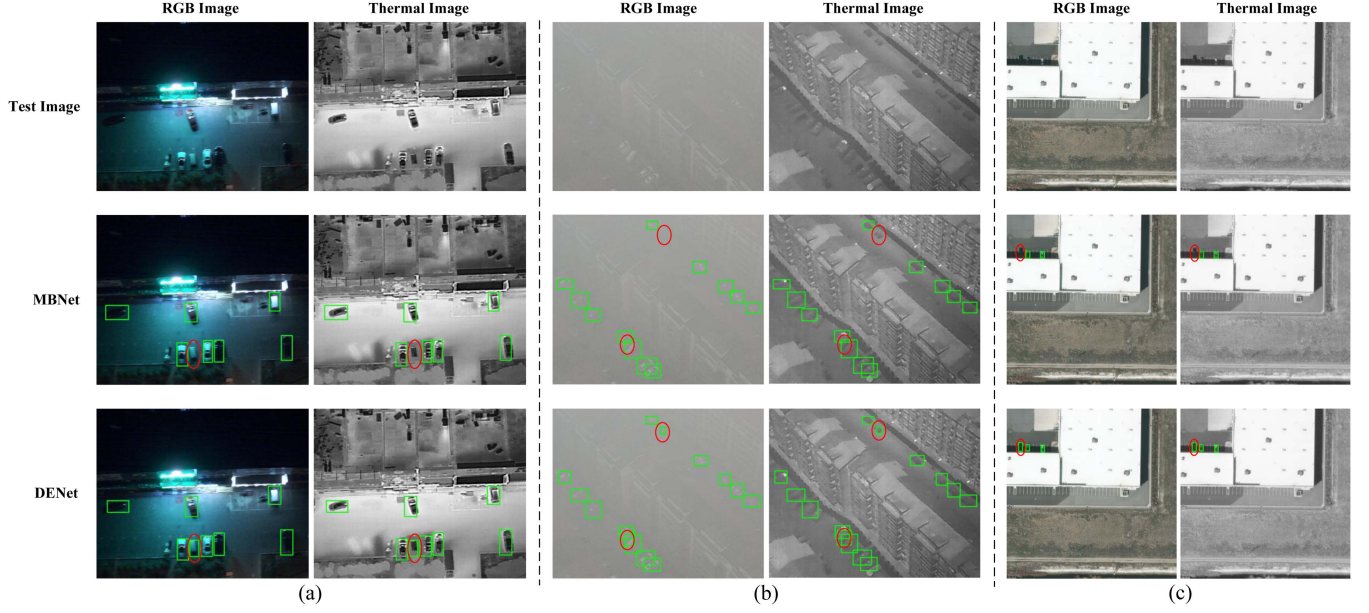


Fig. 5. Visualization results of DENet and the baseline model MBNet on the DroneVehicle and VEDAI datasets. (a) Shows the detection results in nighttime scenes. (b) Shows the detection results in foggy scenes. (c) Shows the detection results in shadowed scenes. Red circles indicate objects detected by DENet but missed by MBNet.

TABLE II
EXPERIMENTAL RESULTS ON VEDAI REMOTE SENSING DATASET

Methods	Modality			mAP(%)
	Visible	Thermal	Visible+Thermal	
YOLO v3 [39]	✓	×	×	62.45
Faster R-CNN [40]	✓	×	×	60.25
RetinaNet [41]	✓	×	×	67.68
DETR [42]	✓	×	×	66.73
YOLO v3 [39]	×	✓	×	64.63
Faster R-CNN [40]	×	✓	×	70.22
RetinaNet [41]	×	✓	×	68.59
DETR [42]	×	✓	×	70.48
UA-CMDet [37]	×	×	✓	73.86
MBNet [43]	×	×	✓	75.52
HyperYOLO [45]	×	×	✓	76.72
DENet (Ours)	×	×	✓	77.83

and thermal images and further applying diffusion enhancement, enabling the model to extract reliable enhanced features.

Similarly, on the VEDAI dataset, we compared our method with several single-modality and multimodal object detection methods, following the same comparison approach as described above. The experimental results are shown in Table II. DENet achieved a mAP(%) score of 83.63%, which is 1.36% higher than MBNet. Among the comparison methods, HyperYOLO, proposed in 2024, achieved the best performance with a mAP(%) of 82.58%, while DENet exceeded it by 1.05%. This improvement is likely due to the high similarity between RGB and thermal images in the VEDAI dataset, where simply fusing multimodal images may introduce redundant information that affects the model's detection performance. However, our proposed RISM and KSIM effectively suppress redundant information between

modalities and balance the information weights across modalities, enabling efficient utilization of information from different modalities. The experimental results demonstrate that our method performs exceptionally well across most datasets.

D. Visualization Results

To facilitate observation, we visualized the results on the test set. Fig. 5 shows the visualization results of MBNet and DENet on the DroneVehicle and VEDAI datasets. From the figure, we can see that DENet's detection performance is superior to MBNet, as it can accurately detect objects with almost no missed detections. As shown in Fig. 5(b), some objects are obscured by dense fog and are no longer visible in the RGB image, while the thermal image lacks sufficient texture information, making it difficult for MBNet to determine whether there are objects in these regions. However, thanks to the DEM in DENet, missing information in these areas can be recovered and enhanced by leveraging multimodal image information, thereby improving the model's ability to detect objects under extreme conditions.

Even though both MBNet and DENet can detect objects in the same test images, it is evident that the bounding boxes regressed by DENet fit the objects more closely than those from MBNet. This is because redundant information often leads to feature blurring and expansion of the target area, resulting in inaccurate bounding boxes, as observed in MBNet. On the other hand, RISM suppresses redundant and irrelevant features before fusing RGB and thermal features, reducing the interference of overlapping features. It retains the unique information related to the object from each modality, preventing the object boundary from becoming blurred or enlarged, ensuring that the regressed bounding boxes are more precise.



Fig. 6. Visualization of several images from the SA-DroneVehicle and EA-DroneVehicle datasets. (a) Shows noisy images caused by sensor anomalies in SA-DroneVehicle. (b) Shows blurry images caused by sensor anomalies in SA-DroneVehicle. (c) Shows images simulating nighttime scenes in the EA-DroneVehicle. (d) Shows images simulating foggy scenes in the EA-DroneVehicle.

TABLE III

ABLATION EXPERIMENTAL RESULTS ON DRONEVEHICLE REMOTE SENSING DATASET

Baseline	RISM	DEM	KSIM	mAP(%)
✓				76.19
✓	✓			76.82
✓		✓		77.53
✓	✓	✓		77.85
✓	✓	✓	✓	78.42

TABLE IV

ABLATION EXPERIMENTAL RESULTS ON VEDAI REMOTE SENSING DATASET

Baseline	RISM	DEM	KSIM	mAP(%)
✓				75.52
✓	✓			76.53
✓		✓		75.46
✓	✓	✓		77.39
✓	✓	✓	✓	77.83

TABLE V

EXPERIMENTAL RESULTS ON SA-DRONEVEHICLE AND EA-DRONEVEHICLE REMOTE SENSING DATASETS

Model	mAP (%)	
	SA-DroneVehicle	EA-DroneVehicle
MBNet	55.95	58.38
DENet	63.14	67.72

For the VEDAI dataset, we adopted the same approach. After integrating RISM, the mAP(%) improved by 1.01% compared to the baseline. Adding DEM on top of this resulted in a 1.87% improvement over the baseline model. Finally, adding KSIM further improved the mAP(%) by 0.44% compared to the “Baseline+RISM+DEM” model.

V. DISCUSSION

Although the DroneVehicle dataset contains some modality failure images, the number of such images is minimal compared to normal images. In real-world scenarios, such as outdoor environments, border areas with no lighting at night, or extreme weather conditions such as heavy fog or sandstorms, modality failure images tend to appear more frequently in large numbers. Therefore, training and testing with only the original dataset is insufficient to evaluate the model’s ability to handle a large-scale modality failure situation. To address this issue, we applied image degradation processing based on the DroneVehicle dataset. Specifically, we first extracted some RGB-T image pairs from the DroneVehicle dataset and randomly added noise, blur, or other distortions to one of the modalities in the image pair to simulate modality-failure scenarios caused by sensor anomalies. This portion of the dataset is referred to as the Sensor Anomaly DroneVehicle (SA-DroneVehicle) dataset, with some example images shown in Fig. 6(a) and (b). Furthermore, we simulated modality failure caused by natural phenomena such as heavy fog or extreme lighting based on the original DroneVehicle dataset. Considering the sensitivity of RGB images to lighting conditions, we only applied image degradation operations to the RGB images in the RGB-T image pairs. This portion of the

E. Ablation Study

To validate the effectiveness of the proposed modules, we conducted ablation experiments on the DroneVehicle and VEDAI datasets, with the results shown in Tables III and IV, respectively. The baseline model we used was MBNet. For the DroneVehicle dataset, starting with the MBNet model, we first integrated the RISM module, resulting in a 0.63% improvement in mAP (%). This indicates that RISM effectively suppresses redundant information during the feature extraction phase, alleviating the interference of redundant information during the feature fusion phase. After adding the DEM module on top of this, the mAP (%) increased by 1.66% compared to the baseline. This is because, in the DroneVehicle dataset, some images suffer from modality failure due to fog or extreme lighting conditions, and DEM can restore these images through the reverse diffusion process. For images without modality failure, DEM also enhances the features. Finally, incorporating KSIM into the model improved the mAP(%) by an additional 0.57% compared to the “Baseline+RISM+DEM” model.

dataset is referred to as the Environmental Anomaly DroneVehicle (EA-DroneVehicle) dataset, with some example images shown in Fig. 6(c) and (d).

We conducted experiments based on these two datasets to verify the performance of our model when faced with large-scale modality failure images. The performance of our proposed DENet and the baseline model MBNet on the two datasets is shown in Table V. From the results, it can be seen that the baseline model MBNet achieved mAP (%) scores of 55.95% and 58.38% on the SA-DroneVehicle and EA-DroneVehicle datasets, respectively, whereas DENet achieved an mAP (%) of 63.14% on the SA-DroneVehicle dataset, which is a 7.19% improvement over MBNet. On the EA-DroneVehicle dataset, DENet achieved an mAP (%) of 67.72%, an improvement of 9.34% compared to MBNet. The experimental results demonstrate that even when dealing with large-scale modality failure images, DENet is still able to perform well due to the DEM module. In addition, we observed that both models performed better on the EA-DroneVehicle dataset compared to the SA-DroneVehicle dataset. This may be because the modality failure images in the EA-DroneVehicle dataset still retain boundary and texture features of the objects, while the modality failure images in the SA-DroneVehicle dataset have lost most of these features due to noise and blur, making it difficult for the models to extract useful information for accurate prediction.

VI. CONCLUSION

To address the issue of modality failure in multimodal object detection, we propose the DENet network, with diffusion enhancement module effectively restores lost image features caused by sensor anomalies, extreme environments, or modality failure. This is particularly important in scenarios where key information is missing. In addition, the RISM reduces redundant RGB and thermal features by minimizing mutual information and incorporating contrastive loss, ensuring complementarity between modalities and strengthening the model's expressive capability in multimodal fusion. Furthermore, the KSIM adaptively adjusts the contribution ratio of RGB and thermal features according to illumination conditions, maintaining balanced and accurate detection results even in conditions of significant lighting variations.

Experimental results show that DENet outperforms existing methods on the DroneVehicle and VEDAI datasets, demonstrating robustness and high detection accuracy on modality-failure images even after applying image degradation, thus validating its stability in complex scenarios. Ablation studies on both datasets further confirm that the proposed modules effectively address issues of modality failure and information redundancy. However, DENet still has certain limitations, such as potentially insufficient recovery capability in extreme cases of complete modality loss, as well as relatively high computational cost. Future research will focus on more refined information recovery techniques and low-cost multimodal fusion mechanisms to further enhance the model's applicability and efficiency.

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